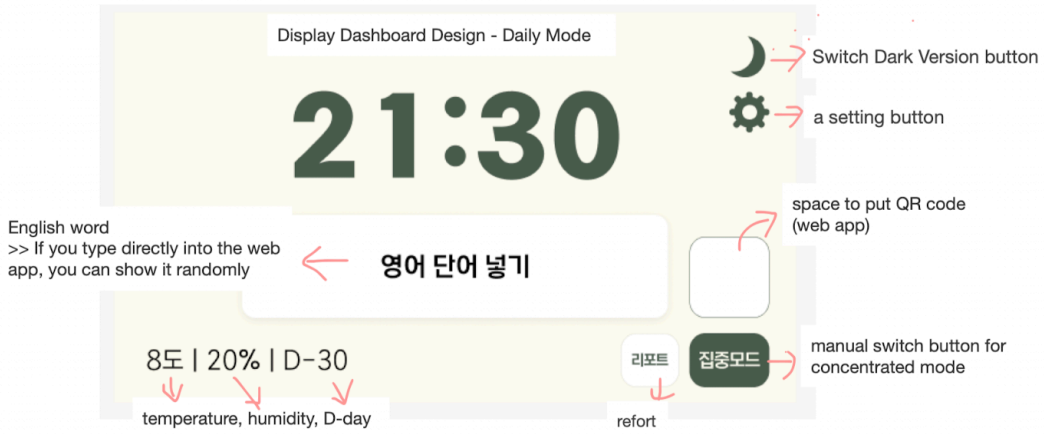
First of all, the scenario is as follows.
I'm thinking of putting the device on the desk. If the user is not detected, the dashboard remains in everyday mode.



[Photo 1] Daily mode

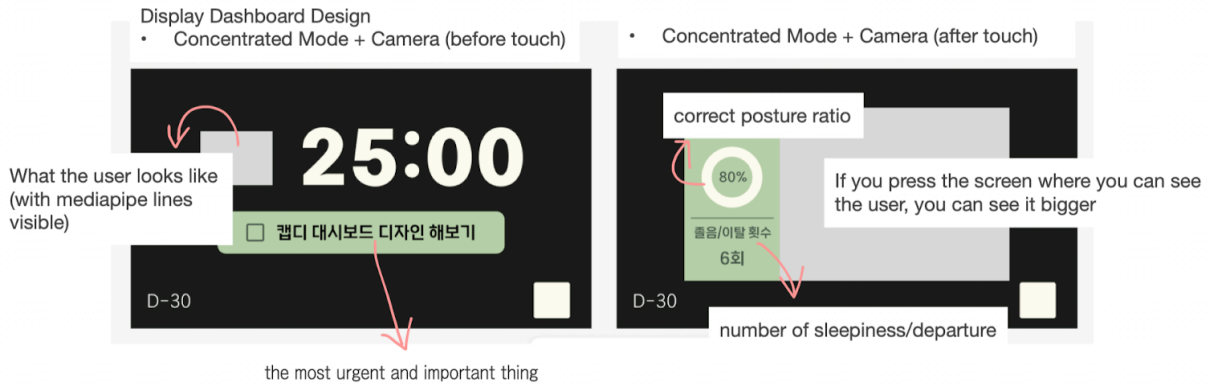
In the daily mode as shown in [Photo 1], you can see the current time, temperature and humidity, D-Day, and what the user wants to memorize. The condition for switching to concentrated mode is (PIR=1 AND ultrasonic sensor<=50cm), or when the user directly presses concentrated mode on the screen. This is when the use of the camera begins. After that, the camera will continue to be used when using intensive mode.



[Photo 2] Setting the user's correct posture criteria

Set the point that serves as the basis for the user's correct posture when the concentrated mode is executed for the first time (first time). (If you want to change the standard of your posture, you can go into the settings and reset them.)

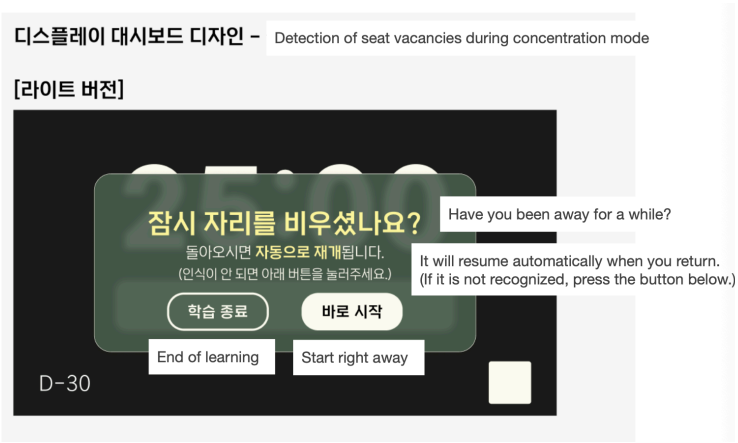
When the posture setting is completed, the screen switches to the concentration mode with the concentration time set by the user in the web app.[Photo 3] (The correct posture ratio and number of sleepiness/departure may change.)



[Photo 3] Focus mode

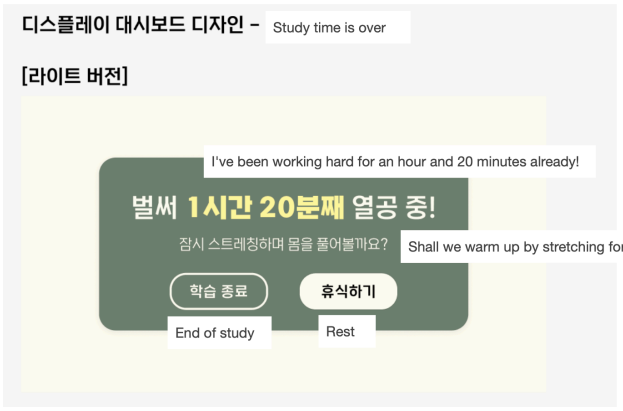
The condition in which the seat space is detected is (PIR=0 AND ultrasound>50cm) or when the user presses the pause button. When a vacancy is detected, a pop-up as shown in Figure 4 will appear and the study timer will stop immediately. If the user is not detected for 20 seconds, pop-up will be displayed, and if the user is detected within 20 seconds, turn off the pop-up and start the timer again. Alternatively, when the user returns to his seat and presses the Start button immediately, the timer starts again.

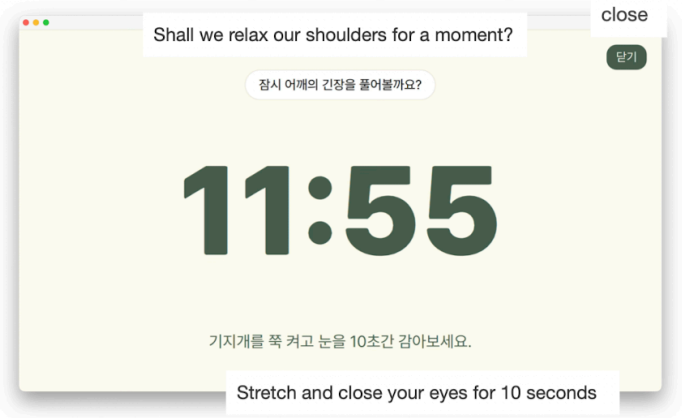
** In ambiguous sections, it remains in its previous state.
ex. Ultrasound<=50cm but PIR=0: Reading still



[Photo 4] Empty pop-up

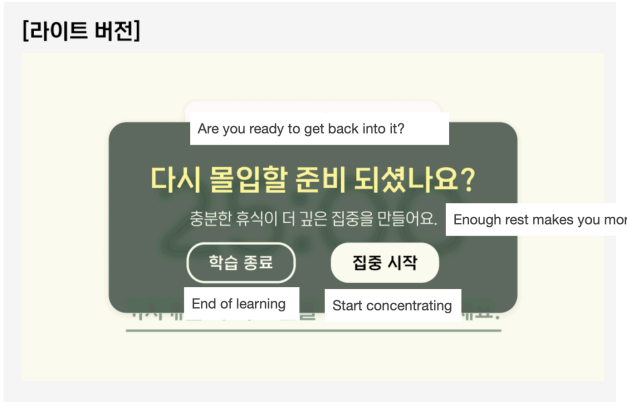
When the timer ends or the user presses the end of learning in the pop-up, a pop-up with the time you studied today appears as shown in [Photo 5]. Press End of Learning to return to the daily mode screen and press Rest to display a screen as shown in [Photo 6]. The camera is not used at this time. Press the Close button to return to the everyday mode screen.





[Photo 5] Ending the concentration mode

[Photo 6] Rest mode

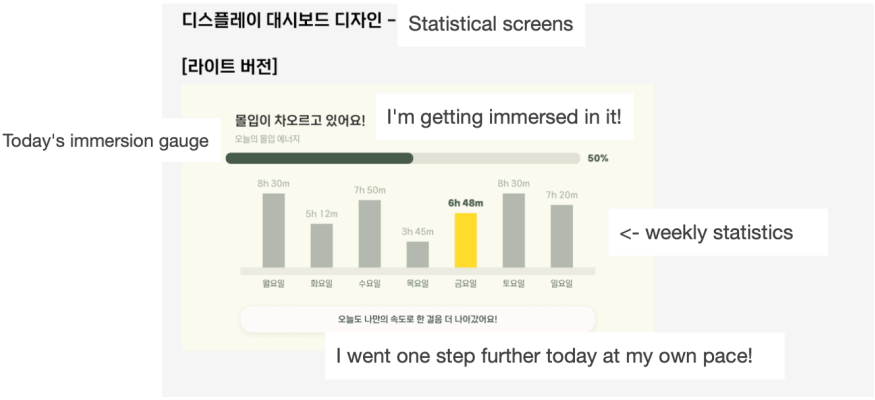


>> [Photo 7] Exits the rest mode

If the break is over without pressing the close button, a pop-up will appear as shown in [Photo 7]. Press End of Learning to return to the daily mode screen [Photo 1] and press Start of Concentration to return to the concentration mode screen [Photo 3].

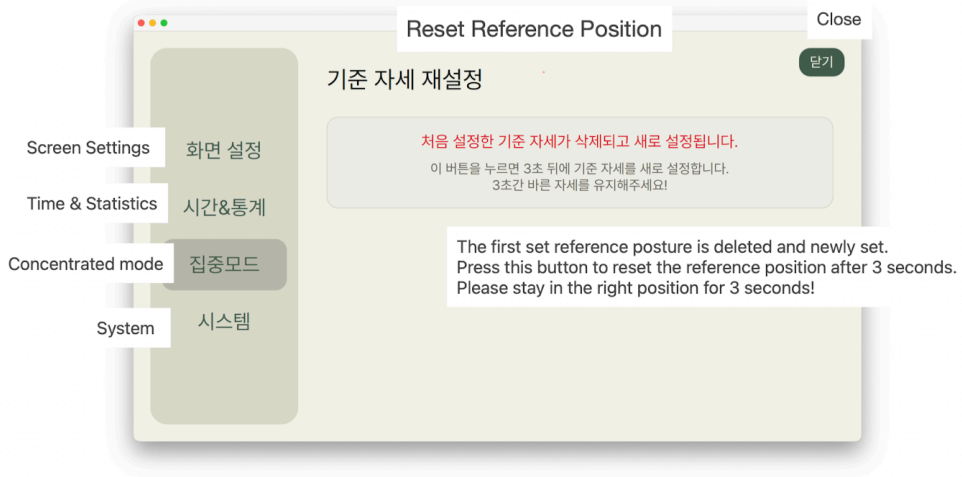
If the screen as shown in [Photo 4] lasts for more than 30 minutes, the data up to this point will be stored and automatically switched to the daily mode. When returning to the daily mode, if the condition of switching to the concentrated mode (PIR=1 AND ultrasound <=50cm) lasts for 10 seconds, the concentrated mode switching pop-up will appear again.If the concentrated mode transition pop-up appears and the Start button is not pressed within 30 seconds, the pop-up disappears and becomes a daily mode screen again. As you study, this process repeats over and over again.

Press the report button on the daily mode screen [Photo 1] to view the report as shown in [Photo 8]. In the dashboard web app, you can see the immersion score of the day and the study time you had for a week. The details will be covered on the web app.



[Photo 8] Report

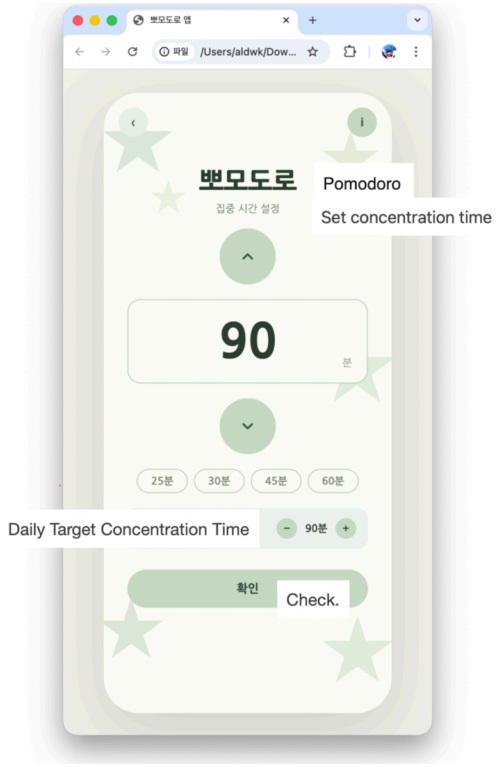
I want to set the pose that is the standard for the correct posture once in the beginning, but if I want to move or set it anew, I plan to make it separately so that it can be reset in the system setting. [Photo 9]



[Photo 9] Reset the reference position

** When not in use for a long time, we plan to prevent overload by turning off the display and camera and only turning on the PIR and ultrasonic sensors to recognize users. But actually, I've also thought about turning off the power by making a system shutdown button in the settings.

This is the area of the dashboard, and the web app allows you to set the target study time for that day and the current study time.[Picture 10] You can set up your to-do through the Eisenhower Matrix [Photo 11] and provide a report on what you have studied.[Photo 12-15]

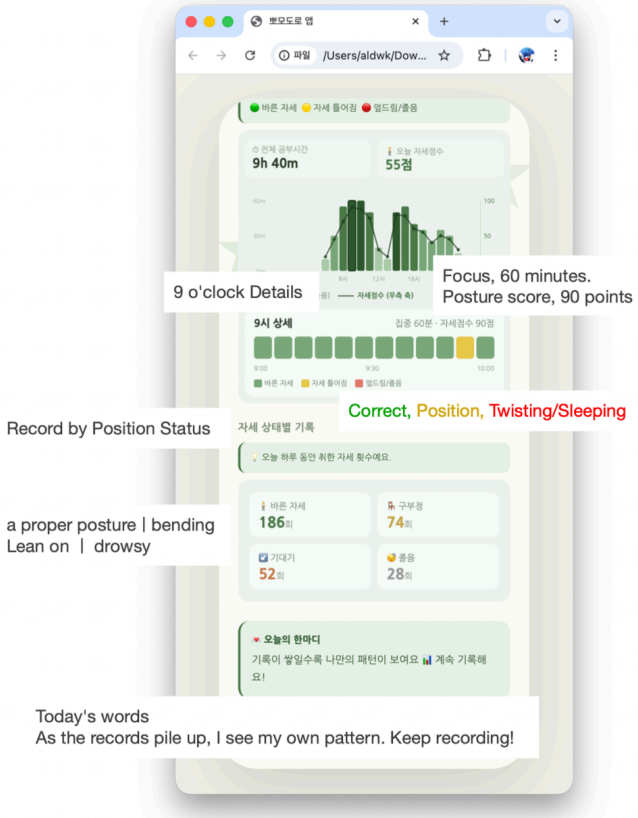


[Photo 10] Pomodoro Timer

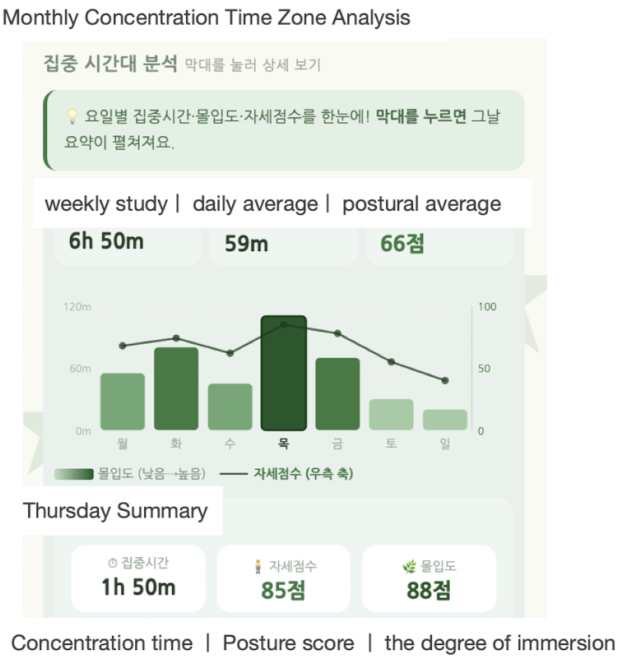


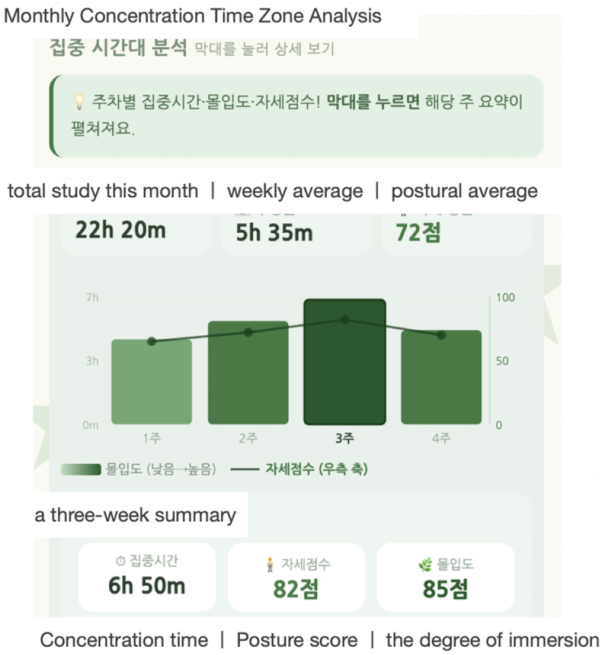
[Photo 11] Eisenhower Matrix





[Photo 12] Study report example data (daily)

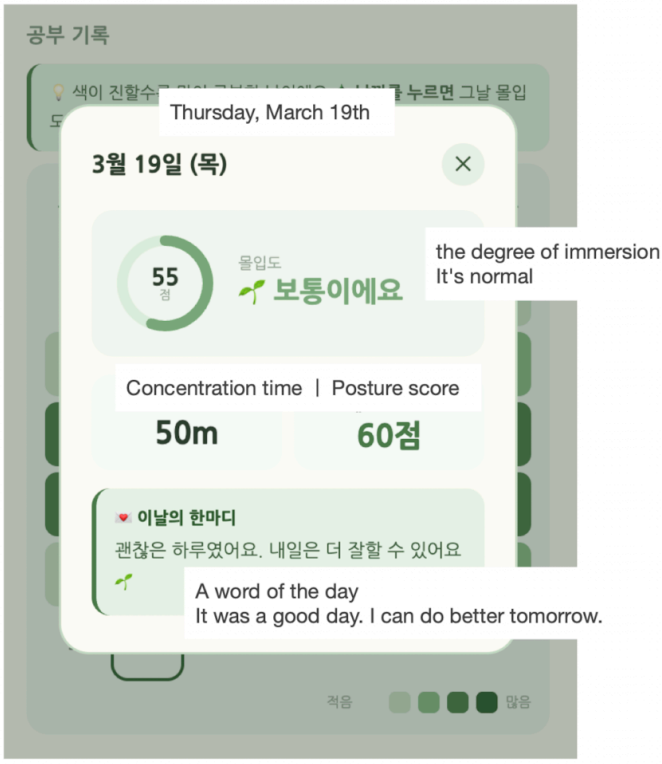




[Photo 13] Study report example data (weekly)

[Photo 14] Study report example data (monthly)





[Photo 15] Study report example data (Monthly study record)

First of all, I'm thinking about this, and here's the algorithm for the camera and the sensors we need.

〈 camera 〉
Use openCV, MediaPipe Face Mesh, MediaPipe Pose
Wrong posture = Less concentration

Posture Type	Model	Method	Threshold
a prone position	Face mesh	Face detection failure	Not detected for more than 20 seconds
drowsy	Face mesh	EAR Average Calculation	EAR<0.20, 8 seconds or more
a bent posture	Pose	Shoulder tilt angle	More than 15 degrees. More than 10 seconds
backrest recline	Pose	Shoulder Y-coordinate descent	More than 30px, more than 10 seconds

** The threshold is an initial value and requires adjustment after hardware testing.

Data to store by posture
1. Total number of times this posture has been taken
2. Duration of all postures
3. Individual Listing - Initial Position Detection Time, corresponding Position Duration

〈 a temperature and humidity sensor 〉
It does not affect the study timer. It just provides information in everyday mode.
** I'm still thinking about whether to display the comfort level or not. It would be nice to mark it, but it's not a priority.

〈 PIR sensor 〉
true - motion detection, false - no motion

〈 an ultrasonic sensor 〉
Measurement <= 50 cm - sitting in front of the desk
Measurement > 50 cm - not in front of the desk
** This value may be modified after testing.

Here are the sensor noise filter values.

〈 an ultrasonic sensor 〉
The value can be greatly splashed due to wall reflections, etc. (ex. 45cm -> 120cm -> 46cm)
Solution: Use the median filter of the last five values, the median ignores one bouncing value.

〈 PIR sensor 〉
Even if you have a user, you can output a false instantaneously.
Solution: If an empty seat is detected, pop up and determine if the user is empty for 20 seconds.

〈 a temperature and humidity sensor 〉
a physically impossible range of values indoors
Solution: Ditch the corresponding value and retain the previous value. (Temperature: -10 degrees to 60 degrees / Humidity: 0% to 100%)

After the design, the dashboard was implemented as a .qml file and called signal.Run the .py file. The web app was implemented as an html file after design. As soon as the current web app work is completed, we will work with the dashboard to check if it works well.